

Project Design Phase-II Technology Stack (Architecture & Stack)

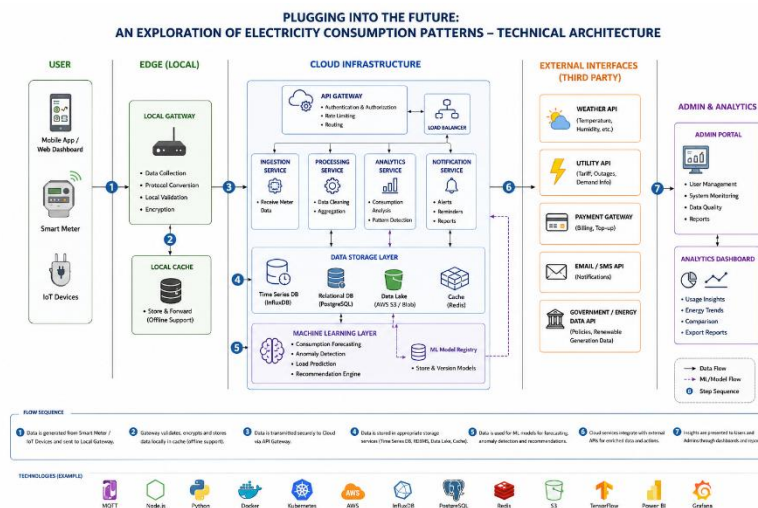
Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all application processes (Data Collection, Processing, Analytics & Reporting).
- Clearly separate **Local/Edge** and **Cloud** infrastructure components.
- Indicate external interfaces (Smart Meter API, Weather API, Utility API, Notification Services).
- Show data storage services (Time-Series Database, Relational Database, Data Lake, Cache).
- Include the Machine Learning module for consumption forecasting, anomaly detection, and energy-saving recommendations.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web/mobile interface for users to monitor electricity consumption, view reports, and receive alerts.	HTML5, CSS3, JavaScript, React.js / Angular, Bootstrap
2.	Application Logic-1	User authentication, registration, login, and profile management.	Python (Flask/Django) or Java (Spring Boot)
3.	Application Logic-2	Electricity data collection, preprocessing, and consumption analysis.	Python, Pandas, NumPy
4.	Application Logic-3	Dashboard generation, report creation, notifications, and energy-saving recommendations.	Flask/Django REST API, Node.js (optional)
5.	Database	Stores user profiles, electricity consumption records, and system configurations.	MySQL / PostgreSQL
6.	Cloud Database	Cloud storage for smart meter data, analytics results, and backups.	AWS RDS, Firebase Firestore, MongoDB Atlas
7.	File Storage	Stores reports, graphs, datasets, and exported PDF/Excel files.	AWS S3 / Google Cloud Storage / Local File System
8.	External API-1	Retrieves weather information to analyze its effect on electricity consumption.	OpenWeather API
9.	External API-2	Fetches electricity tariff, smart meter, or utility provider information.	Utility Provider API / Smart Meter API
10.	Machine Learning Model	Predicts electricity consumption, detects anomalies, and recommends energy-saving actions.	Scikit-learn, TensorFlow, XGBoost
11.	Infrastructure (Server / Cloud)	Hosts the application, APIs, databases, and ML services in cloud or local environment.ssssss	AWS EC2, Docker, Kubernetes, Nginx, Ubuntu Linux

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used for developing the web application, APIs, dashboards, and machine learning models.	React.js, Bootstrap, Flask/Django, Node.js, TensorFlow, Scikit-learn

S.No	Characteristics	Description	Technology
2.	Security Implementations	Secure user authentication, role-based access, encrypted communication, secure API access, and database protection.	JWT Authentication, OAuth 2.0, HTTPS/SSL, SHA-256 Password Hashing, AES Encryption, Firewall, OWASP Security Practices
3.	Scalable Architecture	Three-tier architecture with separate presentation, application, and database layers. Supports cloud deployment and horizontal scaling for increasing users and smart devices.	3-Tier Architecture, Docker, Kubernetes, REST APIs, AWS Auto Scaling
4.	Availability	Ensures high availability through cloud deployment, backup services, load balancing, and continuous monitoring for uninterrupted access.	AWS EC2, Elastic Load Balancer (ELB), AWS CloudWatch, RDS Backup, Multi-AZ Deployment
5.	Performance	Optimized for fast processing of electricity consumption data using caching, efficient databases, asynchronous processing, and optimized APIs.	Redis Cache, PostgreSQL/MySQL Indexing, Nginx, REST APIs, CDN (CloudFront), Pandas for Data Processing

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>